Satria Wibawa Hadiwijaya

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Graduate electrical engineer with Honours, from a double degree program. Started with UAV power and signal wiring on a university robotics team, then went deeper into control and instrumentation through analogue electronics, power systems and embedded firmware. Most recently the control engineer on a student haemodialysis machine. Drew its full electrical schematic and wrote the MicroPython firmware that catches a fault mid-cycle. Currently looking for a graduate opportunity to begin a career in control and instrumentation across power and renewable energy. The end goal is to become a specialist deep enough in that discipline to advance the technology itself, and to bring AI into everyday engineering work.

Education

Monash University — *Clayton, VIC, Australia*
Bachelor of Electrical and Computer Systems Engineering (Honours) — *Jul 2024 – Jun 2026*

- WAM 75.2 / 100, completed June 2026 through the UI–Monash International Double Degree (KKI) program.
- High Distinctions in Engineering Integrated Design (89), Final Year Project (88), Professional Practice (86), Autonomous Vehicle Systems (85) and Solar Energy (83).
- Coursework: Control Systems Design, Electrical Energy Systems, Solar Energy, Hydropower, Analogue Electronics, Medical Instrumentation, Information and Networks, Engineering Electromagnetics.

Universitas Indonesia — *Depok, Indonesia*
Bachelor of Electrical Engineering — *Aug 2022 – Jun 2026*

- GPA 3.67 / 4.0, Cum Laude. Coursework: Electrical Power Engineering, Control Engineering, Embedded Systems, Electronic Circuits, Algorithms and Programming.

Experience

Monash Young MedTech Innovators (MYMI) — *Melbourne, VIC*
Control Engineer, Project Nephros — *Sep 2025 – Jun 2026*

- Produced the schematic the team now works from, covering all five subsystems of a student haemodialysis unit in Fritzing: thermistors, pressure and flow sensors, stepper drivers, pumps and the LCD.
- Put live blood and dialysate temperature in front of operators on two thermistor channels, by programming the sensing and LCD readout onto an Arduino Nano ESP32 in MicroPython.
- Moved fault detection from post-run to mid-cycle with MicroPython firmware that polls the sensors on a fixed cycle, smooths the signal, and flags any out-of-range temperature or pressure.
- Carried the build through to a working machine exhibited at two showcases, by selecting every component, holding the bill of materials, and settling electronics, fluid-circuit and enclosure interfaces across three sub-teams.

Georg Fischer Indonesia — *Karawang, Indonesia*
Human Resources Administrative Assistant Intern — *Jan 2026 – Feb 2026*

- Migrated payroll from SunFish to BIPO HRMS with no records lost, by preparing the full 2026 employee roster and cross-checking every division against the old system before cutover.
- Processed insurance claims, payslips and time-and-attendance records each month, and cleared discrepancies directly with the staff concerned.
- Gathered the evidence behind a new decentralised roster that around 60% of staff rated above the old one, through structured rostering and overtime meetings with every subdivision.
- Cut about 15% off the department's weekly reporting time by pairing AI tooling with Excel techniques from its internal training course, the first AI used anywhere in that cycle.

Society of Monash Electrical Engineers (SMEE) — *Melbourne, VIC*
Marketing Director; Industry & Technical Sub-Committees — *Nov 2024 – Jun 2026*

- Converted 40+ sponsorship approaches into signed partners including Gridwise and Aurora Energy, and drew 100+ students to Tech Industry Night alongside the Australian energy sector.
- Delivered seven technical workshops through 2025 covering LTspice, C programming, lab skills, Bluetooth, motors, Arduino and sensors, and solar panels, and specified the components and bill of materials for each.
- Ran the SMEE Annual Robot Building Competition for 100+ participants as technical supervisor across both days, plus the month of build workshops leading into it.

Universitas Indonesia Robotics Team — *Depok, Indonesia*
Electrical and Autonomous Vehicle Systems Division — *Feb 2023 – Nov 2024*

- Cut telemetry dropouts by roughly 20% on quad- and hexacopter platforms, and steadied them in flight, by rebuilding battery placement, servo integration, and the wiring between flight controller, ESCs and receiver.
- Brought the incoming electrical division up to build standard through structured modules on power architecture, component selection, and Pixhawk and ROS-based VTOL integration.
- Contributed to the V-1 Raptor VTOL that reached the KRTI 2023 national finals against 25 of Indonesia's top universities.

Projects

Swingbus: Power-System Benchmarking Toolkit (Grid Plotter) *Sep 2026 – Present*

Personal project, v0.9.5 beta: React, ECharts, Electron and Python

- Built an offline toolkit for benchmarking power-system simulations against field measurements, an easier alternative to Enerplot. The Grid Plotter overlays traces with A/B cursors, linked zoom and one tab per test case; a Grid Calculator sizes SMIB tests (Zs from SCR and X/R, source voltage, fault impedance for a dip). Ships as one HTML file and a Windows app.
- Reverse-engineered the binary PSCAD `.psout` and PSS®E `.out`/`.outx` formats from raw bytes with no vendor libraries, plus MATLAB and COMTRADE readers; Python and in-app readers matched on all 373 columns of real grid-code test files.
- Opened a 1 GB, 14.5-million-row recording in about 18 s through a background worker and spike-preserving min/max thinning; covered by 529 unit tests.

Image Processing for a Bionic Eye: Retinal Encoder Evaluation *Jul 2025 – Jun 2026*

Final Year Project (Honours), graded 88, High Distinction

- Built a retinal encoder and phosphene-rendering pipeline in Python with Pulse2Percept. The model covers photoreceptor filtering, centre-surround receptive fields, contrast gain control and ON/OFF pathways, and turns camera frames into simulated bionic-eye percepts.
- Designed and ran a 15-participant perceptual study under Monash ethics approval MUHREC 44457: four encoder configurations, three custom recognition tasks, 60 randomised trials.
- Found spatial resolution to be the dominant driver of recognition accuracy. The best configuration reached 69.2% mean accuracy against 55.1% for the coarsest, a 14-point gap between a 625- and a 144-electrode array.

Wildlife Observation Robot: Visual Detection and GUI *Aug 2025 – Nov 2025*

ECE4191 Electrical Design Project, Team E08

- Trained a YOLOv12 animal detector on 4,000+ images in Google Colab. It resolves a detection inside 10 seconds across an 80-degree field of view at 30–40 cm.
- Tuned the live video pipeline (TCP streaming, JPEG decoding, PyTorch inference) to hold 10 FPS analysis under 300 ms end-to-end latency.
- Wired the detector into a custom Tkinter GUI with bounding boxes and detection history, and took the interface from 5 FPS to 20 FPS.
- Ran the full vision and GUI stack in live demonstration: 21 automatic detections against a target of 8.

True RMS-to-DC Converter (Analogue Circuit Design) *Aug 2025 – Nov 2025*

ECE3161 Mini-Project, Team B22

- Designed the multiplier, divider and square-root stages, and ran LTspice sweeps to check RMS accuracy from DC to 20 kHz.
- Built and tested several implementations of the RMS core, matching transistor and resistor networks for stable current-mode operation.
- Co-developed and debugged the precision rectifier so it held accuracy on large DC offsets and low-amplitude signals.
- Laid out and troubleshot the wiring so measurements repeated to within 3% error.

Deep Learning Traffic Sign Recognition Model *Feb 2025 – May 2025*

CIV4100 Autonomous Vehicle Systems Project

- Cleaned and augmented a balanced 60,000+ image set across 43 traffic-sign classes.
- Trained a custom CNN to 97.1% validation accuracy, then applied AlexNet transfer learning to reach 98.7% and lift Kaggle test accuracy from 67.4% to 71.1%.
- Probed robustness with metamorphic tests and isolated translation as the weakest axis.

Real-Time Video Processing System (DE-10 FPGA) *Feb 2025 – May 2025*

ECE3073 Computer Systems, Team C15

- Gave the board hands-free control by integrating its accelerometer, so a tap, double-tap or tilt switches display mode or rotates the quad-display layout.
- Held gesture-to-response latency to 0.1–0.2 s across four real-time video-processing modes.

Technical Skills

Programming: Python, JavaScript (React), C, C++, MicroPython, Verilog, MATLAB/Simulink.
Hardware and Design: PCB and schematic (KiCad, Proteus, Fritzing), embedded systems (ESP32, Arduino, Pixhawk), LTspice.
Tools and Platforms: PyTorch, Google Colab, Git, Factory I/O, EcoStruxure, QGroundControl, ROS.
Engineering Domains: Control engineering, instrumentation, power systems, industrial automation, embedded firmware.
AI and Automation: Agentic workflow design, LLM-assisted documentation and data processing, prompt engineering.
Languages: English (native, PTE Academic 90/90), Bahasa Indonesia (native).

Certifications

- **PTE Academic** (Jul 2026): overall 90 / 90, with 90 / 90 in each of Listening, Reading, Speaking and Writing.
- **Monash Engineering CPD** (2024–2026): 340 certified hours, 220 of them engineering, across MYMI and SMEE.